

Motivation Letter

UNESCO and AI-Lab/CBPF Remote Access to Artificial Intelligence Infrastructure

I am applying for remote access to the UNESCO and AI-Lab/CBPF artificial intelligence infrastructure to advance my PhD research at Lviv Polytechnic National University on a practical question that I repeatedly encounter in bioinformatics: why do classical machine learning methods so often outperform, match, or remain preferable to deep learning and foundation-model approaches on biological tasks?

My professional background is in applied machine learning and language technologies. At WebSpellChecker LLC I have worked on custom transformer-based architectures for grammatical error correction, data preprocessing, model training, and reproducible training pipelines. In parallel, my doctoral work focuses on integrative causal-oriented analysis of molecular biological markers for prioritizing anti-aging interventions. This combination has made the gap between modern AI infrastructure and biological data practice especially visible to me. Many biological datasets are small, noisy, heterogeneous, batch-affected, and expensive to validate. In these conditions, the best model is not always the largest model, and stronger benchmarking is needed before complex architectures are treated as the default.

The proposed project, "Understanding Why Classical ML Outperforms Deep LLMs in Bioinformatics," is motivated by this tension. I want to build a transparent benchmark and analysis workflow that compares classical ML methods, deep neural networks, and bio-foundational models across representative bioinformatics tasks. The goal is not to reject deep learning, but to identify when it provides genuine value, when simpler models are more robust, and which measurable properties of biological data explain the difference.

I have already made progress toward this direction by assembling and screening literature on biological foundation models, preserving decision logs, building a model catalogue, and experimenting with document-processing workflows for full-text review. The next step is to connect this evidence base to executable benchmarks where model behavior can be tested under matched data splits, metrics, compute budgets, and reproducibility controls.

Access to remote AI infrastructure is important because the project requires controlled experiments across multiple model families, datasets, random seeds, and preprocessing variants. On local resources, it is difficult to separate scientific conclusions from compute limitations. With stable GPU and CPU infrastructure, I can run systematic comparisons, keep reproducible logs, test sensitivity to data size and feature structure, and document negative as well as positive results. This is essential for a fair comparison between classical and deep approaches.

My expected output is an open, reproducible benchmarking package with clear experiment logs, model-selection guidance, and a data-centric framework for deciding when bioinformatics problems benefit from foundation models. I am particularly motivated to make this work useful for researchers who do not have unlimited compute budgets and who need reliable, interpretable methods for biological discovery. The UNESCO/AI-Lab infrastructure would directly support this objective by giving the project the computational depth required to move from anecdotal comparisons to systematic evidence.